

Unrestricted Lower Bounds for Rational Binary Balance Languages

Exact State Complexity for Deterministic Pushdown Recognition of $L_{p,q}$

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Abstract

For coprime integers $p, q \geq 1$, consider the rational binary balance language

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The problem studied here grew out of a $p:q$ generalization of an earlier completion-aware 2:1 deterministic pushdown construction. That generalization isolates the signed residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w)$$

as the semantic coordinate governing continuations and right quotients, and it raises a representation-independent question: is the state cost visible in compact residual realizations forced by the language itself?

We prove that it is. Every deterministic ε -pushdown automaton recognizing $L_{p,q}$ in the regular-target model satisfies

$$|Q| \geq \max(p, q).$$

The lower bound is unrestricted with respect to the physical pushdown representation: no canonical residual encoding, homogeneous-tail hypothesis, bounded push length, or residual-faithful configuration map is assumed.

The proof combines quotient rigidity with protected stack pumping. A unary pump produces a family of common buried cuts carrying p synchronized residue phases. If a phase eventually exposes the common cut, different phases must return to it in different states. If exposure is deferred forever, a second independent unary pump appears above the protected cut. The two pumps generate a two-parameter family of reachable stacks whose acceptance, after one fixed continuation, is exactly a linear diagonal

$$c q k = p d \ell.$$

Fixed-continuation predecessor stack languages are regular, while this diagonal is not, yielding a contradiction. Hence all p phases expose and $|Q| \geq p$; the symmetric argument gives $|Q| \geq q$.

We also give a self-contained ε -free right-endmarker construction with $|Q| = \max(p, q)$ and a three-symbol stack alphabet. Therefore, under the right-endmarker convention,

$$Q_{\min}(L_{p,q}) = \max(p, q).$$

1 Introduction

The family

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}$$

is a simple commutative family whose membership is controlled by one integer imbalance. Its pushdown realizations, however, need not be simple. The same semantic imbalance may be

distributed among finite control, the visible stack top, deep stack structure, and transient ε -normalization.

The present lower-bound problem arose naturally from the $p:q$ generalization of an earlier 2:1 completion-aware deterministic pushdown construction. In the original special case, the stack could be read operationally as a record of the imbalance still to be cancelled by future input. The generalization separates this operational picture from the semantic invariant

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w),$$

under which the two input letters act by translations

$$r \xrightarrow{1} r + q, \quad r \xrightarrow{0} r - p.$$

This residual viewpoint leads to compact deterministic pushdown realizations and, more importantly for the present paper, to the question of whether the observed finite-control cost is an artifact of those realizations or an intrinsic property of $L_{p,q}$.

Figure 1 records the historical 2:1 origin. The figure is motivational only: no transition rule, state organization, stack encoding, or completion convention from that machine is assumed in the lower-bound proof.

The conceptual progression leading to the present theorem is summarized in Figure 2. The point is not that every later recognizer resembles the original machine; the point is precisely the opposite. The $p:q$ residual is semantically rigid while its physical pushdown lifts can vary substantially. The present theorem removes the representation hypothesis entirely.

Main result. For coprime p, q , we prove

$$\boxed{|Q| \geq \max(p, q)}$$

for every deterministic ε -pushdown recognizer of $L_{p,q}$ in the model specified below. The lower bound is independent of the stack alphabet size, the maximum length of stack replacement words, the presence of deterministic ε -moves, and the way in which the machine encodes the residual.

The proof does not attempt to reconstruct a canonical residual representation inside an arbitrary competitor. Instead, it combines two representation-independent facts. First, the residual classifies right quotients, so equal reachable physical configurations must carry equal residuals. Second, suffix-minimum positions on a live unary computation yield context-independent protected stack pumps. These pumps produce synchronized residue phases over one family of buried cuts.

Every phase then falls into one of two cases. If the cut is exposed, a congruence argument forces state separation. If the cut remains buried forever, the deferred computation produces a second independent pump. The two pumps form a rectangular family of reachable stacks. After one fixed continuation, correctness requires acceptance exactly on an infinite linear diagonal. Standard pushdown predecessor regularity makes the same parameter set regular, a contradiction.

The existing state hierarchy for deterministic pushdown automata is classical [1, 3]; our claim is not a new hierarchy theorem for DPDAs in general. The question here fixes the natural commutative family $L_{p,q}$ and asks for its exact parameter-dependent state cost over arbitrary deterministic pushdown representations. Related work on stateless and fixed-state pushdown-alphabet hierarchies studies a different resource axis [2, 3]. The regularity theorem used in the deferred branch is classical pushdown reachability machinery [4].

Organization. Section 2 fixes the model and the residual quotient. Section 3 proves unary nonrecurrence and height escape. Section 4 derives the protected pump and synchronized phases. Section 5 proves the exposure/deferred dichotomy and eliminates permanent deferral. Section 6 gives the unrestricted lower bound. Section 7 gives a matching three-symbol construction, and Section 8 derives exact state complexity.

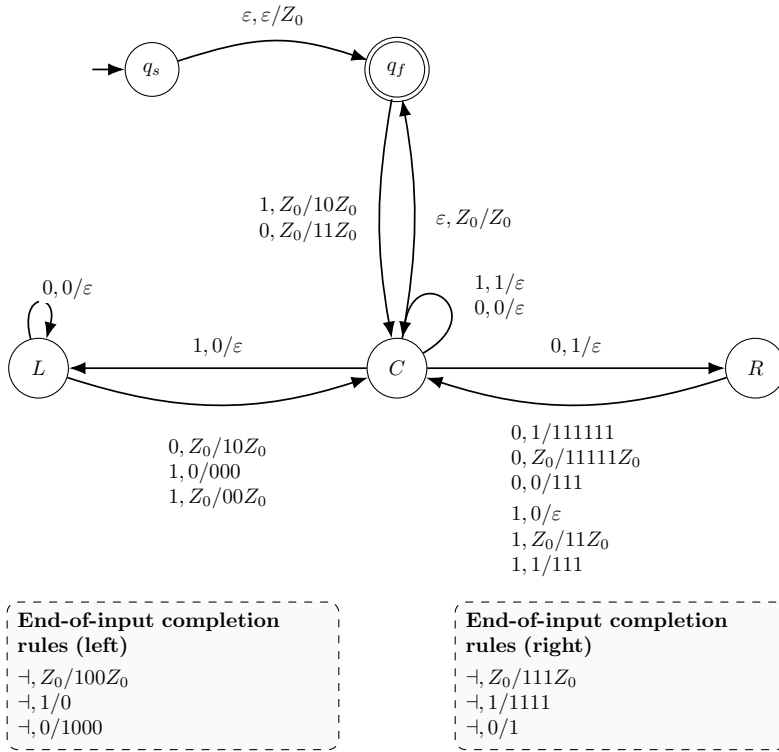


Figure 1: A formalized version of the completion-aware 2:1 DPDA that motivated the residual/completion viewpoint. The center/left/right recognition logic is shown together with the end-of-input completion callouts. This figure records the origin of the research program; the unrestricted lower-bound proof below assumes none of its transition rules or physical encoding choices.

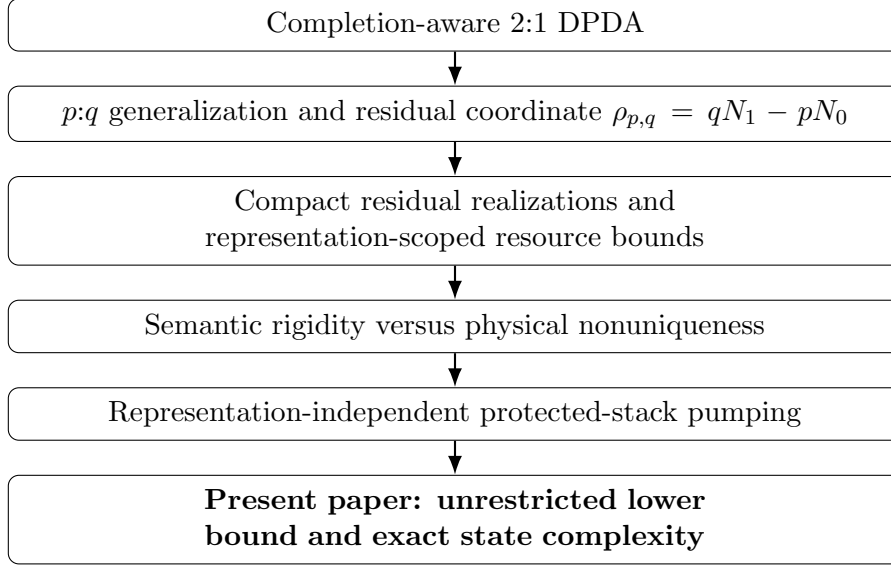


Figure 2: Research progression leading to the present result. The 2:1 machine supplied the completion viewpoint; the $p:q$ generalization isolated the semantic residual; subsequent resource questions exposed the distinction between representation-dependent optimality and language-intrinsic necessity. The present paper resolves the latter problem.

2 Model and semantic quotient

2.1 Pushdown model

A deterministic pushdown automaton has finite control Q , finite stack alphabet Γ , and transitions that inspect the current state, the visible top stack symbol, and either one input symbol or ε . The stack top is written on the right. Stack words, including regular languages of stacks used later, are therefore written and read from bottom to top. Thus a transition

$$\delta(q, X, a) = (q', \beta)$$

acts as

$$(q, \eta X, av) \vdash (q', \eta \beta, v).$$

A single transition may replace X by an arbitrary fixed finite word $\beta \in \Gamma^*$. Determinism means that for each state/top pair there is at most one move on each input letter, and if an ε -move is enabled then no input-consuming move is enabled at that state/top pair.

For the lower bound we use a *regular-target* acceptance convention. Fix a regular set $T \subseteq Q \times \Gamma^*$ of accepting control–stack configurations. After the finite binary input has been consumed, the word is accepted exactly when the deterministic post-input computation reaches T without consuming any further binary input. Reaching T in zero post-input steps is allowed. Ordinary final-state, empty-stack, and final-state-plus-empty-stack acceptance are obtained by the corresponding regular choices of T . A mandatory right-endmarker convention is handled by treating the endmarker as the final symbol of the continuation. The fixed-continuation regularity consequence needed later is proved separately in Theorem 5.2; it is not part of the model definition.

The matching upper bound in Section 7 is stated explicitly for a mandatory right endmarker.

Remark 2.1 (Meaning of “unrestricted”). *The lower-bound opponent may use an arbitrary finite stack alphabet, arbitrary finite replacement words, deterministic ε -moves, noncanonical intermediate configurations, and any physical encoding compatible with correct recognition. No raw-conjugacy, homogeneous-tail, or canonical normal-form assumption is imposed.*

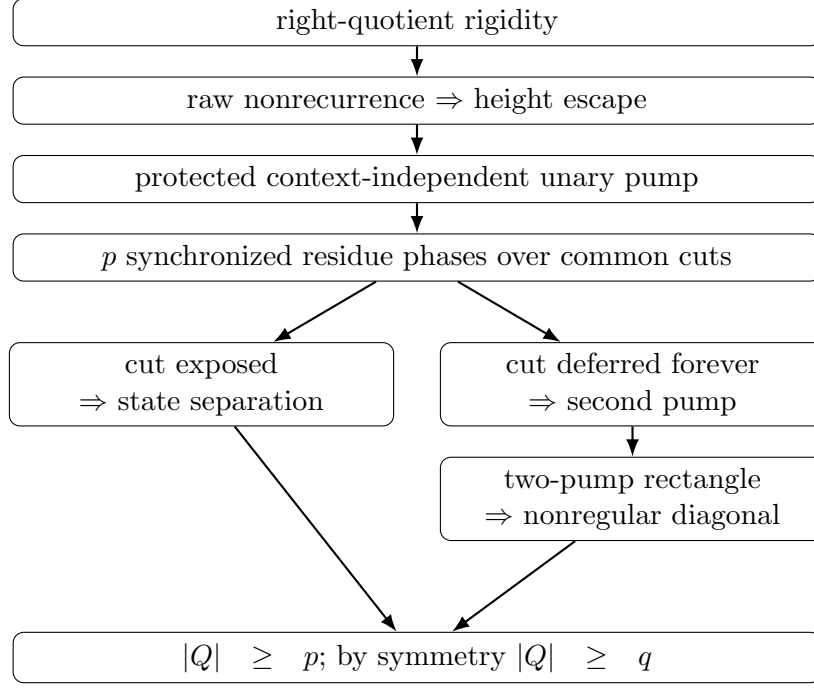


Figure 3: Architecture of the unrestricted lower-bound proof. Exposure charges synchronized residue phases directly to finite control; permanent deferral instead creates an independent second pump, which is ruled out by fixed-continuation regularity.

2.2 Residuals

For $w \in \{0, 1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then

$$w \in L_{p,q} \iff \rho(w) = 0,$$

and

$$\rho(wa) = \begin{cases} \rho(w) - p, & a = 0, \\ \rho(w) + q, & a = 1. \end{cases}$$

Lemma 2.2 (Integer residual realizability). *If $\gcd(p, q) = 1$, then for every $t \in \mathbb{Z}$ there exists $z \in \{0, 1\}^*$ with*

$$\rho(z) = t.$$

Proof. Choose integers x_0, y_0 with $qx_0 - py_0 = t$. All integer solutions are

$$x = x_0 + pk, \quad y = y_0 + qk.$$

For sufficiently large k , both x, y are nonnegative. Any binary word with $N_1 = x$ and $N_0 = y$ then has residual t . $\square$

Lemma 2.3 (Residual classification of right quotients). *For all $u, v \in \{0, 1\}^*$,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z ,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 2.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ while

$$\rho(vz) = \rho(v) - \rho(u) \neq 0,$$

so $vz \notin L_{p,q}$. □

Lemma 2.4 (Physical configurations determine residuals). *If two consumed prefixes u, v lead to the same reachable physical configuration of a deterministic recognizer of $L_{p,q}$, then*

$$\rho(u) = \rho(v).$$

Consequently every reachable physical configuration C has a well-defined semantic label $\Theta(C) \in \mathbb{Z}$.

Proof. Starting from one and the same deterministic physical configuration, the set of accepting continuations is identical. Thus

$$u^{-1}L_{p,q} = v^{-1}L_{p,q},$$

and Theorem 2.3 gives $\rho(u) = \rho(v)$. □

For every reachable input-consuming transition,

$$\Theta \xrightarrow{1} \Theta + q, \quad \Theta \xrightarrow{0} \Theta - p,$$

while ε -moves preserve Θ .

3 Unary nonrecurrence and height escape

Consider the computation induced by the infinite unary input 1^ω . For every $n \geq 0$, both 1^n and 1^{n+1} have accepting finite extensions: their residuals nq and $(n+1)q$ admit cancelling continuations by Theorem 2.2. Consequently, after consuming 1^n the deterministic machine cannot halt, reject permanently, or become trapped in an infinite ε -computation before consuming the next 1; otherwise no extension beginning with that next 1 could ever be accepted. Hence the unary computation consumes arbitrarily many input symbols, and every finite consumed unary prefix is live.

Lemma 3.1 (Raw nonrecurrence on the positive unary ray). *No full raw configuration occurs twice on the live computation over 1^ω .*

Proof. Suppose the same configuration C occurs twice. If the two occurrences have consumed different numbers $m < n$ of input symbols, then Theorem 2.4 assigns C both residuals mq and nq , impossible.

If the two occurrences have consumed the same number of input symbols, the intervening segment consists entirely of ε -moves. Determinism then repeats the same ε -cycle forever, so the next input symbol can never be read. This contradicts liveness of the current unary prefix. □

Corollary 3.2 (Unary height escape). *Along the computation over 1^ω , stack height eventually leaves every fixed finite bound.*

Proof. For each fixed height bound H , there are only finitely many state/stack configurations with height at most H . If infinitely many raw positions remained in that bounded region, some full configuration would repeat, contradicting Theorem 3.1. □

Definition 3.3 (Suffix minimum). A raw position C_i on an infinite computation is a suffix minimum if

$$h(C_i) \leq h(C_t) \quad \text{for every } t \geq i.$$

The following elementary consequence will be used repeatedly.

Lemma 3.4 (Unbounded suffix minima). *If an integer-valued height sequence tends to infinity, then it has suffix-minimum positions at arbitrarily large heights.*

Proof. For every raw position u , the tail $\{h(t) : t \geq u\}$ has a minimum because $h(t) \rightarrow \infty$. Choose a position at which this tail minimum is attained. As u tends to infinity, the selected positions cannot remain in a finite set, and their heights must become arbitrarily large. $\square$

4 Protected stack pumping and synchronized phases

A nonempty stack configuration $(q, \eta X)$ has *mode* (q, X) .

Lemma 4.1 (Protected unary pump). *There exist a state $r \in Q$, a symbol $X \in \Gamma$, an integer $c \geq 1$, a nonempty word $U \in \Gamma^+$, and a lower context η such that*

$$(r, \eta X) \xRightarrow{1^c} (r, \eta U X),$$

where the displayed macro includes all forced ε -moves and no raw configuration of the macro has stack height below $|\eta X|$.

Moreover the macro is *context-independent*: replacing the untouched lower context η by any other lower context leaves the raw transition sequence unchanged as long as that context is not exposed.

Proof. By Theorem 3.4, suffix-minimum positions occur at arbitrarily large heights. Since only $|Q||\Gamma|$ state/top modes exist, some mode (r, X) occurs at suffix minima of arbitrarily large height. Choose two such positions $t_1 < t_2$ with the same mode and with the later height strictly larger. Write the earlier stack as ηX .

Because t_1 is a suffix minimum, the computation from t_1 to t_2 never drops below height $|\eta X|$. Exposing any symbol of the lower prefix η would first require popping the visible X and reaching height $|\eta| < |\eta X|$. Hence η is never exposed or modified. At t_2 the state and visible top are again r, X , so the larger stack has the form

$$\eta U X$$

for some $U \in \Gamma^+$.

Let c be the number of input symbols 1 consumed between t_1 and t_2 . Every transition of the segment sees only the active suffix above the untouched lower context, so replacing η by any other lower context reproduces the same deterministic raw transition sequence. This proves context independence.

It remains to show $c \geq 1$. If $c = 0$, the segment is a context-independent, height-increasing ε -macro from mode (r, X) back to itself:

$$(r, \eta X) \xRightarrow{\varepsilon} (r, \eta U X).$$

Context independence would then repeat the same forced macro from $(r, \eta U^m X)$ for every $m \geq 0$, producing an infinite deterministic ε -computation and preventing the next symbol of the live unary input from ever being consumed. This is impossible. Therefore $c \geq 1$. $\square$

4.1 Iteration and protected cuts

By context independence, for every $k \geq 0$,

$$(r, \eta U^k X) \xRightarrow{1^c} (r, \eta U^{k+1} X).$$

Each copied raw segment is identical to the original segment above its lower context, and hence never goes below the height of its own starting configuration. Therefore the cuts

$$H_k = \eta U^k$$

are protected throughout the entire future computation on 1^ω starting at

$$C_k = (r, H_k X).$$

After discarding finitely many initial k and rebasing the index, we may assume every H_k is nonempty.

This explicit repetition argument is important: we do not infer protection of the synthetic endpoints merely from the fact that the original two selected positions were suffix minima.

Lemma 4.2 (Synchronized p -phase family). *For each $j \in \{0, \dots, p-1\}$ there exist a state q_j and a nonempty upper stack word W_j , independent of k , with the following property. Starting from C_k , consume exactly j additional symbols 1 and then take every forced ε -move until the computation is again ready to inspect the next unread input symbol. The resulting raw configuration is*

$$D_{k,j} = (q_j, H_k W_j).$$

There is a constant $A \in \mathbb{Z}$ such that

$$\Theta(D_{k,j}) = A + kcq + jq.$$

Proof. The future 1^ω -computation from every C_k remains above H_k . Hence the relative computation above the cut depends only on the common state/top data and not on k . Liveness excludes an infinite ε -divergence after the j th consumed symbol, so the stated next-input boundary is well defined. For fixed j its state and upper stack are therefore independent of k ; call them q_j, W_j . The upper word is nonempty because exposing H_k would contradict protection of the cut along the 1^ω future. The residual formula follows because each pump block contributes cq and each of the additional j symbols contributes q . $\square$

5 Exposure, deferral, and the diagonal obstruction

Fix a phase j and follow the unique computation on 0^ω from each $D_{k,j}$. Until the common buried cut H_k is exposed, the machine sees only the state and the stack portion above H_k . Thus the relative raw computation is independent of k .

We say that H_k is *exposed* when the full stack is exactly H_k at a raw position. This event cannot be skipped downward. If the current stack is $H_k Z$ with a nonempty upper suffix, one PDA transition rewrites only the visible top symbol and therefore lowers total stack height by at most one. Consequently any computation that ever reaches below the height of H_k must pass through a raw configuration whose stack is exactly H_k first.

Lemma 5.1 (Uniform exposure dichotomy). *For each phase j , exactly one of the following holds.*

- (i) *There exist $z_j \geq 0$ and $s_j \in Q$ such that for every k the first exposure of H_k occurs after exactly z_j consumed zeros, at configuration*

$$(s_j, H_k).$$

- (ii) *For every k , the computation on 0^ω from $D_{k,j}$ never exposes H_k .*

Proof. Before exposure, every transition acts strictly above H_k . The initial relative configuration (q_j, W_j) is independent of k , so the entire relative raw computation is the same for every k . Either its upper stack becomes empty after one fixed raw history and a fixed number z_j of consumed zeros, or it never becomes empty. In the former case the state at that moment is also fixed. $\square$

5.1 Fixed-continuation regularity

Lemma 5.2 (Fixed-continuation stack regularity). *Fix a control state s and a finite continuation v (including the mandatory right endmarker, when that convention is used). Let*

$$R_{s,v} = \{S \in \Gamma^* : \text{starting from } (s, S), \text{ the fixed continuation } v \text{ leads to acceptance}\}.$$

Then $R_{s,v}$ is regular.

Proof. Write $v = v_1 \cdots v_m$. Augment the finite control by an input-position coordinate $i \in \{0, \dots, m\}$. Original ε -moves leave i fixed; at position $i < m$, only a transition consuming the required letter v_{i+1} advances the coordinate to $i + 1$. Acceptance of the fixed word v is therefore reduced to predecessor reachability of a regular target set in an ordinary pushdown system. If one uses the standard bounded-push presentation of pushdown systems, each transition that writes a longer fixed word is compiled into finitely many auxiliary control states; this normalization does not change the predecessor set on the original configurations. Regular sets of pushdown configurations are effectively closed under predecessor reachability by the standard saturation theorem [4]. Restricting the resulting regular configuration set to the fixed control state s yields a regular language of stacks. $\square$

Lemma 5.3 (Deferred exposure is impossible). *No phase $j \in \{0, \dots, p-1\}$ can satisfy alternative (ii) of Theorem 5.1.*

Proof. Assume phase j is deferred. Fix one k . The 0^ω -computation remains strictly above H_k . Every finite consumed zero-prefix is live, because its current residual has a finite cancelling continuation by Theorem 2.2. Hence the computation consumes arbitrarily many zeros and cannot become trapped in an ε -cycle.

No full raw configuration can repeat: a repetition after different numbers of consumed zeros would assign one physical configuration two residuals differing by a nonzero multiple of p , while a repetition at the same consumed input position would create a deterministic ε -cycle. Thus the relative upper-stack height above H_k escapes every finite bound.

Apply Theorem 4.1, with 0 in place of 1, to this relative zero computation. There exist a state s , a visible symbol Y , an integer $d \geq 1$, a nonempty word $V \in \Gamma^+$, and a fixed upper word β such that

$$(s, H_k \beta Y) \xrightarrow{0^d} (s, H_k \beta V Y).$$

Because the complete relative zero computation before exposure is identical for every k , the same second pump appears above every H_k . Iterating it gives reachable configurations

$$E_{k,\ell} = (s, \eta U^k \beta V^\ell Y), \quad k, \ell \geq 0.$$

There is a constant $B \in \mathbb{Z}$ such that

$$\Theta(E_{k,\ell}) = B + c q k - p d \ell.$$

Choose, using Theorem 2.2, a fixed binary word v_0 with $\rho(v_0) = -B$. If the model uses a mandatory right endmarker, append it and write $v = v_0 \dashv$; otherwise put $v = v_0$. Since every $E_{k,\ell}$ is reachable and the automaton recognizes $L_{p,q}$,

$$E_{k,\ell} \xrightarrow{v} \text{accept} \iff c q k = p d \ell.$$

By Theorem 5.2, the stack language $R_{s,v}$ is regular. Hence so is

$$K = \{a^k \# b^\ell : \eta U^k \beta V^\ell Y \in R_{s,v}\}.$$

Indeed, if a DFA recognizes $R_{s,v}$, reading each fixed block η, U, β, V, Y induces a fixed transformation of the DFA state set. A finite automaton on the parameter alphabet $\{a, \#, b\}$ can therefore simulate the transformations corresponding to $\eta U^k \beta V^\ell Y$, proving that K is regular.

On the other hand correctness gives

$$K = \{a^k \# b^\ell : cqk = pd\ell\}.$$

With $g = \gcd(cq, pd)$, set

$$\alpha = \frac{pd}{g}, \quad \beta = \frac{cq}{g}.$$

Then

$$K = \{a^{\alpha n} \# b^{\beta n} : n \geq 0\}.$$

This language is nonregular. Indeed, if N were a pumping length, the word $a^{\alpha N} \# b^{\beta N}$ would have every admissible pumping segment lying wholly in the initial a -block; pumping that segment changes the number of a 's while leaving the number of b 's fixed, destroying the equation $\beta k = \alpha \ell$. This contradiction eliminates the deferred alternative. $\square$

6 The unrestricted lower bound

Theorem 6.1 (Positive-side state lower bound). *Every deterministic ε -pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq p.$$

Proof. By Theorem 5.3, every phase $j \in \{0, \dots, p-1\}$ exposes the common cut. Fix one k . For each phase,

$$D_{k,j} \xrightarrow{0^z_j} (s_j, H_k).$$

If $s_j = s_t$ for distinct j, t , the two exposure configurations are physically identical, because their stacks are both exactly H_k . By Theorem 2.4, their residuals are equal:

$$A + kcq + jq - pz_j = A + kcq + tq - pz_t.$$

Thus

$$q(j - t) = p(z_j - z_t).$$

Since $\gcd(p, q) = 1$, this implies $p \mid (j - t)$, impossible for $0 < |j - t| < p$. Therefore $s_0, \dots, s_{p-1}$ are pairwise distinct and $|Q| \geq p$. $\square$

Theorem 6.2 (Negative-side state lower bound). *Every deterministic ε -pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq q.$$

Proof. Apply the proof of Theorem 6.1 after interchanging the two input letters, interchanging p and q , and replacing the residual by its negative. $\square$

Theorem 6.3 (Unrestricted state lower bound). *For coprime $p, q \geq 1$, every deterministic ε -pushdown recognizer of $L_{p,q}$ in the stated regular-target model satisfies*

$$\boxed{|Q| \geq \max(p, q)}.$$

Proof. Combine Theorems 6.1 and 6.2. $\square$

Remark 6.4. *The theorem is stronger than the product lower bound*

$$|Q||\Gamma| \geq \max(p, q)$$

that one might first seek. The argument charges the ratio parameter directly to finite control, without any restriction on $|\Gamma|$.

7 A matching three-symbol construction

We now give a self-contained upper bound. This section is independent of the lower-bound representation and is included so that the exact state-complexity statement is proved entirely within the present paper.

Assume first that $p \geq q$. Let

$$Q = \{R_0, \dots, R_{p-1}\}, \quad \Gamma = \{A, B, Z_0\}.$$

We encode each residual $r \in \mathbb{Z}$ as follows. For $r \geq 0$, write uniquely

$$r = pk + s, \quad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\zeta(r) = (R_s, Z_0 A^k).$$

For $r < 0$, write uniquely

$$r = u - qh, \quad h \geq 1, \quad 0 \leq u < q,$$

and define

$$\zeta(r) = (R_u, Z_0 B^h).$$

For $0 \leq s < p$, put

$$s^+ = (s + q) \bmod p, \quad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

and

$$u_s = (s - p) \bmod q, \quad h_s = \frac{u_s + p - s}{q} \geq 1.$$

For $0 \leq u < q$, put

$$u^- = (u - p) \bmod q, \quad d_u = \frac{u^- - u + p}{q} \geq 1.$$

With the stack top on the right, define the following binary rules:

$$\begin{aligned} \delta(R_s, A, 0) &= (R_s, \varepsilon), & \delta(R_s, A, 1) &= (R_{s^+}, A^{c_s+1}), \\ \delta(R_s, Z_0, 1) &= (R_{s^+}, Z_0 A^{c_s}), & \delta(R_s, Z_0, 0) &= (R_{u_s}, Z_0 B^{h_s}), \end{aligned}$$

and for $0 \leq u < q$,

$$\delta(R_u, B, 1) = (R_u, \varepsilon), \quad \delta(R_u, B, 0) = (R_{u^-}, B^{d_u+1}).$$

No ε -transition is used.

A mandatory right endmarker $\dashv$ is accepted only from

$$\zeta(0) = (R_0, Z_0).$$

For example, declare R_0 final and define only

$$\delta(R_0, Z_0, \dashv) = (R_0, Z_0)$$

among endmarker moves.

Theorem 7.1 (Three-symbol residual realization). *For every coprime $p, q \geq 1$, there is an ε -free right-endmarker DPDA recognizing $L_{p,q}$ with*

$$|Q| = \max(p, q), \quad |\Gamma| = 3.$$

Proof. Assume $p \geq q$. We verify

$$\delta(\zeta(r), 0) = \zeta(r - p), \quad \delta(\zeta(r), 1) = \zeta(r + q)$$

for every residual r .

Let $r = pk + s \geq 0$. If $k \geq 1$, input 0 pops one A , giving

$$pk + s - p = p(k - 1) + s.$$

For input 1,

$$pk + s + q = p(k + c_s) + s^+,$$

so the rule replacing A by A^{c_s+1} gives exactly $\zeta(r + q)$. When $k = 0$, the 1-rule uses the same carry identity, while

$$s - p = u_s - qh_s,$$

so the boundary 0-rule enters exactly the negative encoding.

Now let $r = u - qh < 0$. Input 1 pops one B . If $h > 1$ this gives $u - q(h - 1)$; if $h = 1$ it exposes Z_0 and leaves $(R_u, Z_0) = \zeta(u)$. For input 0,

$$u - p = u^- - qd_u,$$

and hence

$$r - p = u^- - q(h + d_u),$$

which is implemented by replacing one visible B by B^{d_u+1} .

Therefore every binary input symbol realizes the correct residual translation in one ordinary transition. Starting from $\zeta(0)$, after reading w the machine is in $\zeta(\rho(w))$. The right endmarker is accepted exactly at $\zeta(0)$, so the recognized language is $L_{p,q}$.

If $q > p$, exchange 0 and 1, exchange p and q , exchange A and B , and negate the residual. The resulting machine has $q = \max(p, q)$ states. $\square$

8 Exact state complexity

Theorem 8.1 (Exact unrestricted right-endmarker state complexity). *For coprime $p, q \geq 1$, among all deterministic pushdown automata in the lower-bound model whose inputs are terminated by the mandatory symbol $\dagger$,*

$$\boxed{Q_{\min}(L_{p,q}) = \max(p, q)}$$

under deterministic pushdown recognition with a mandatory right endmarker.

Proof. Theorem 6.3 gives the lower bound and Theorem 7.1 gives the matching upper bound. $\square$

Corollary 8.2 (Arbitrary positive parameters). *For arbitrary $p, q \geq 1$, let $d = \gcd(p, q)$. Since*

$$L_{p,q} = L_{p/d, q/d},$$

the right-endmarker state complexity is

$$\boxed{Q_{\min}(L_{p,q}) = \max\left(\frac{p}{d}, \frac{q}{d}\right)}.$$

9 Related work and positioning

State hierarchies for deterministic pushdown automata are classical. Masopust recalls, with Harrison as the classical reference, that the number of control states establishes an infinite hierarchy within the deterministic context-free languages and that binary witnesses are available [1, 3]. A different descriptive axis is the size of the pushdown alphabet. Meduna, Vrabel, and Zemek established pushdown-alphabet hierarchies for stateless pushdown models [2]; Masopust subsequently obtained binary witnesses for the deterministic stateless case and extended the method to fixed-state realtime DPDAs [3].

The present problem has a different form. We do not construct a separate witness language for each level of a general state hierarchy. Instead, we fix the natural commutative family

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\}$$

and determine its parameter-dependent state minimum over arbitrary deterministic pushdown representations. The lower bound is therefore a representation-independent complexity statement for one arithmetic language family.

The regularity ingredient used in Theorem 5.2 is also classical. Bouajjani, Esparza, and Maler represent regular sets of pushdown configurations by finite automata and compute predecessor sets by saturation [4]. Its role here is specific: if a synchronized phase remains hidden below the visible stack forever, two independent unary pumps arise, and predecessor regularity rules out the nonregular linear acceptance diagonal that those pumps would force.

To the best of our knowledge, the exact unrestricted formula

$$Q_{\min}(L_{p,q}) = \max(p, q)$$

under the right-endmarker convention has not previously been stated for this rational binary balance family. No claim is made that state lower bounds or state hierarchies for deterministic pushdown automata are new in general.

10 Scope and significance

The main lower bound is not an optimality theorem for one residual encoding. The competing automaton is free to hide phase information in deep stack structure, to use large stack alphabets, to push arbitrarily long fixed words, and to insert deterministic ε -normalization. The proof survives these freedoms because it never identifies a canonical physical residual representation.

Its semantic input is only the quotient rigidity of $\rho_{p,q}$. Its physical input is only generic LIFO locality: a protected lower context cannot influence a computation until it is exposed, and one top rewrite cannot skip a buried cut downward. Permanent refusal to expose the cut is itself costly, because it creates the independent second pump that produces the diagonal contradiction.

Thus the lower bound visible in compact $p:q$ realizations is not merely a property of those realizations. The inequality

$$|Q| \geq \max(p, q)$$

is intrinsic in the stated lower-bound model, while the matching equality is asserted specifically under the mandatory right-endmarker convention.

11 Conclusion

We have established an unrestricted state lower bound for the rational binary balance family

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\}.$$

For coprime p, q , every deterministic ε -pushdown recognizer in the stated regular-target model satisfies

$$|Q| \geq \max(p, q).$$

Under the mandatory right-endmarker convention, a self-contained ε -free three-symbol construction attains the bound, and therefore

$$Q_{\min}(L_{p,q}) = \max(p, q).$$

The proof is representation-independent. Right-quotient rigidity supplies the semantic coordinate, protected suffix-minimum pumping supplies synchronized phases, and the exposure/deferred dichotomy forces every phase to become visible in finite control. The decisive deferred branch converts hidden phase information into a two-pump reachable rectangle; fixed continuation regularity then contradicts the nonregular linear acceptance diagonal.

The result completes the transition from the residual structure suggested by the $p:q$ generalization to a statement about arbitrary deterministic pushdown realizations. The quantity $\max(p, q)$ is therefore not an artifact of one normalization scheme or one stack encoding: it is the exact finite-control cost under the right-endmarker convention.

Statements and Declarations

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Ethics Approval and Consent. Not applicable.

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